EVENT BOOKING PLATFORM

A synopsis Submitted
In Partial Fulfillment of the Requirements
for the Degree of

BACHELOR OF TECHNOLOGY
(COMPUTER SCIENCE AND ENGINEERING)

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Declaration

I hereby declare that this SYNOPSIS report entitled, "**EVENT BOOKING PLATFORM**", being submitted by me to the **DR. A P J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW** for the award of the degree of Bachelor of Technology (CSE) is a record of bonafide work carried by me (Roll No.- 2004720109012). I worked under the guidance and supervision of **AMIT KAUSHAL, (HOD, INSTITUTE OF TECHNOLOGY & MANAGEMENT), Maharajganj.**

The matter embodied in this report has **not** been submitted to any other university or institute earlier for the award of any degree or diploma.

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B. TECH. (CSE) VIIITH SEM (2022-23)

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PLACE: **MAHARAJGANJ**

DATE: 23/04/2023

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Introduction

Event Booking Platform is a website for discovering events and booking tickets online. It is built with **PHP, Bootstrap, CSS and JavaScript** on a **MySQL** database. Organisers (partners) create events with one or more dates and define ticket types with their own price and quantity. Visitors browse events by city, category, artist and place, and registered users book tickets for themselves and their guests. Payments are made online through the **Razorpay** API and SMS alerts are sent through the **Fast2SMS** API. Every guest receives a unique ticket code, which organiser staff verify at the gate to check the guest in or deny entry.

There is a centralized database in which all the information is managed. The administrator approves events, verifies partner KYC details, manages categories, artists, places and content, and processes payouts to organisers. Organisers can promote events by buying boost packages, and users and partners can raise support tickets or report events and receive notifications. The website involves four types of users / modules: Administrator, Partner (Event Organizer), User and Guest.

Objectives

- To automate ticket booking so that users can find an event, select tickets and pay online without standing in queues.
- To give organisers one place to publish events, set ticket types and prices, and track bookings and sales.
- To provide secure online payment through Razorpay and to issue a unique ticket code for every guest.
- To keep users informed through SMS (Fast2SMS), notifications and support tickets.
- To help organisers manage entry at the venue and promote events through boost packages.
- To reduce time and manual effort for organisers and administrators.

Roles and Responsibilities

- Analysis of user, partner and administrator modules for the Event Booking Platform
- Designed mock up UI with Bootstrap, CSS and JavaScript for all modules
- Implemented the event, ticket booking, payment, SMS and check-in modules in PHP
- Designed the MySQL tables w.r.t. users, events, tickets, payments and payouts
- Integrated the Razorpay and Fast2SMS APIs; fixed issues raised in review by the project guide

Requirements Analysis

The requirement analysis is done in order to understand the problem the software system is to be solved. The emphasis is on identifying what is needed for the system, not how it will achieve its goal. The goal of requirement activities is to document requirements in a software requirement specification. There are two major phases: (1) problem understanding or analysis and (2) requirement specification.

In problem analysis the aim is to understand the problem and its context. Once the requirements are understood, they must be specified in a requirement specification document covering all functions, performance, input and output formats and design constraints. We can define a requirement as "A condition of capability by a user to solve a problem or achieve an objective", or "A condition or a capability that must be met or possessed by a system…. to satisfy a contract, standard, specification, or other formally imposed document."

Software Specifications

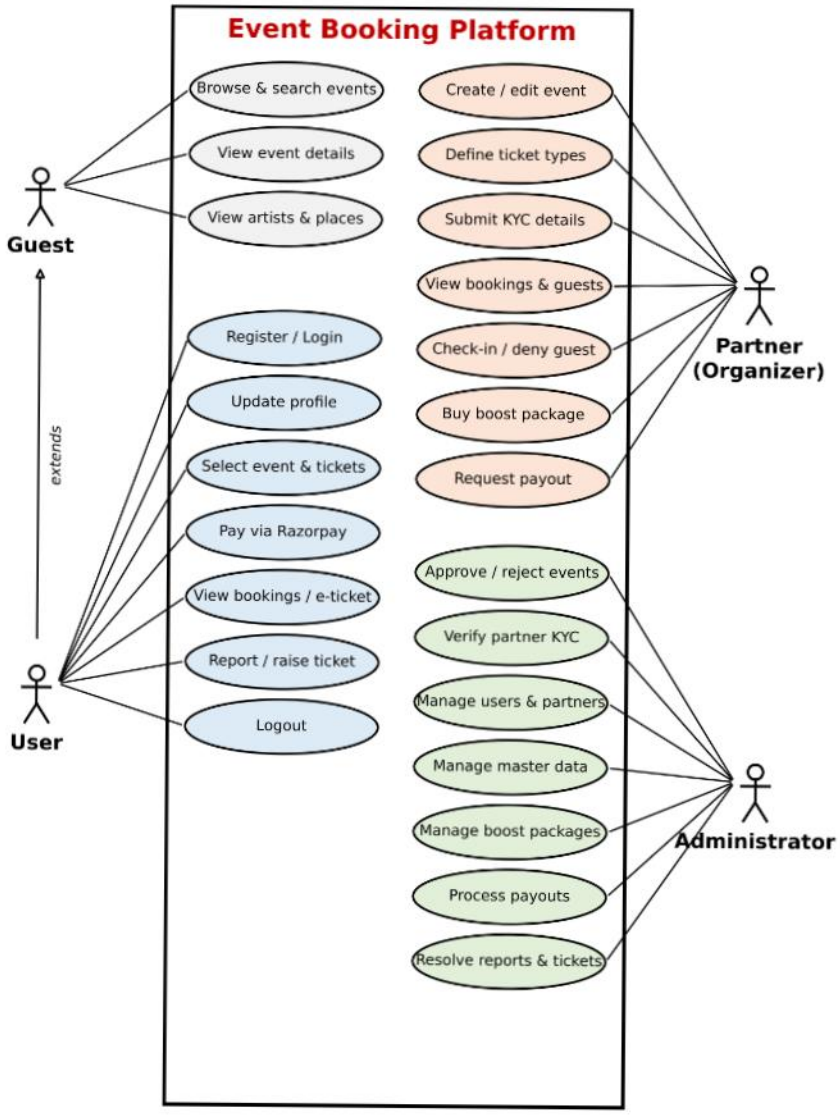
Database	MySQL
Server side Languages & other tools	PHP, REST APIs
Client Side Languages & Other tools	HTML, CSS, Bootstrap, JavaScript
Payment Gateway API	Razorpay
SMS API	Fast2SMS

System Design

System design is the process or art of defining the architecture, component, module, interface the data for a system to specified requirement. Object-oriented analysis and design (OOAD) methods are becoming the most widely used method for computer system design. The UML has become the standard language used in Object-oriented analysis and design.

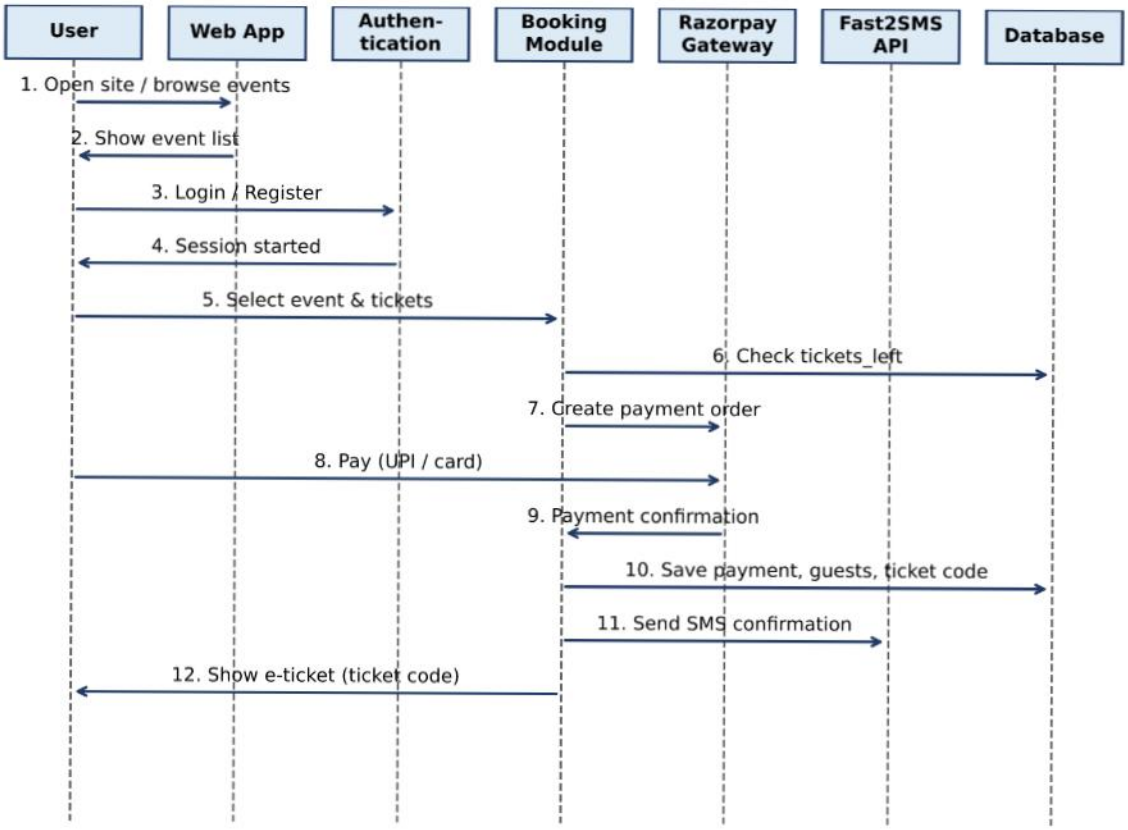
Use Case Diagram

A use case diagram is a representation of a user's interaction with the system that shows the relationship between the user and the different use cases in which the user is involved.



Sequence Diagram

A sequence diagram is an interaction diagram that shows how processes operate with one another and in what order. The diagram below shows the flow of booking tickets, from login to payment, SMS confirmation and e-ticket.

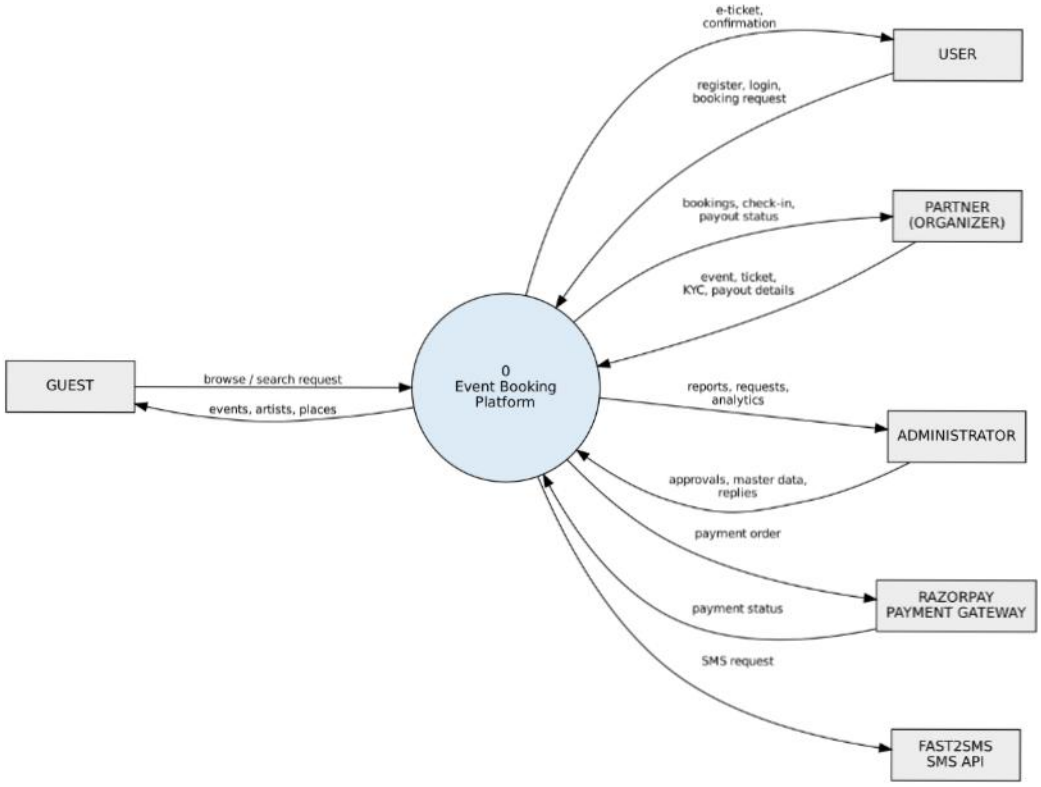


Data Flow Diagram

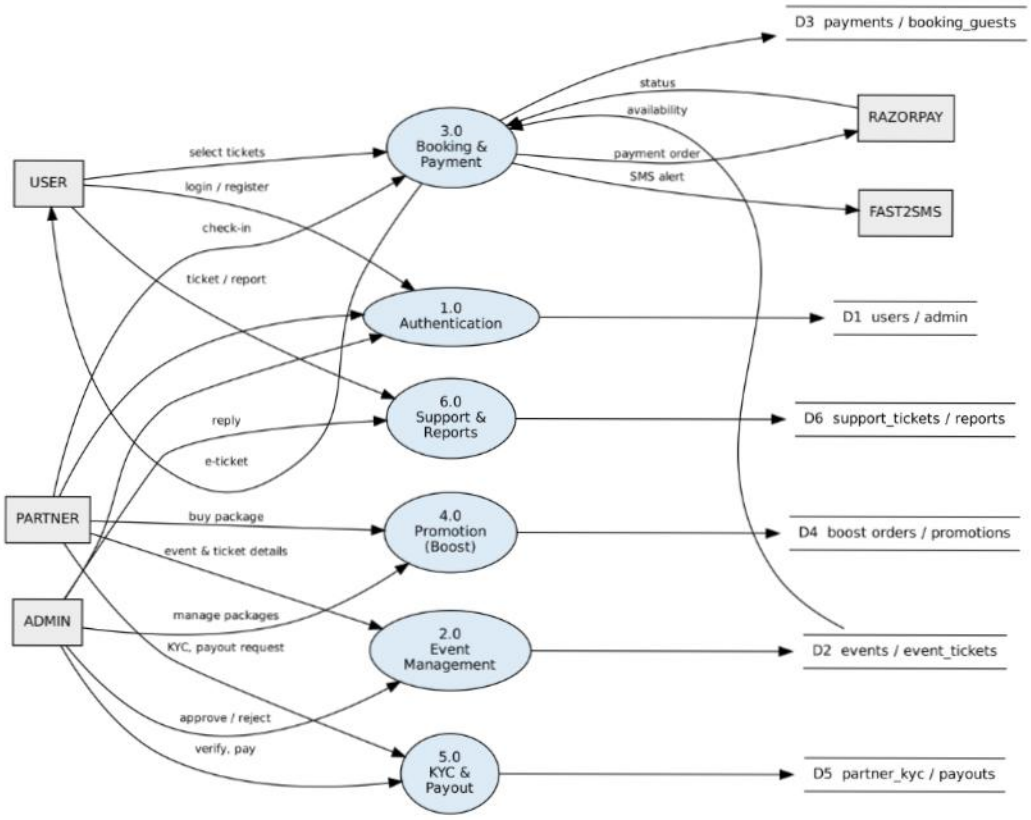
A DFD (bubble chart) clarifies system requirements and identifies the major transformations that become programs in system design. Bubbles (circles) represent processes that transform data, lines with arrowheads represent data flows, squares represent external entities that are the source or destination of data, and open-ended narrow rectangles represent data stores. A DFD describes what data flows rather than how it is processed.

- Name and number every process; show the direction of flow from top to bottom and left to right.
- Names of data stores, sources and destinations are written in capital letters; arrows should not cross each other.

Level 0 DFD

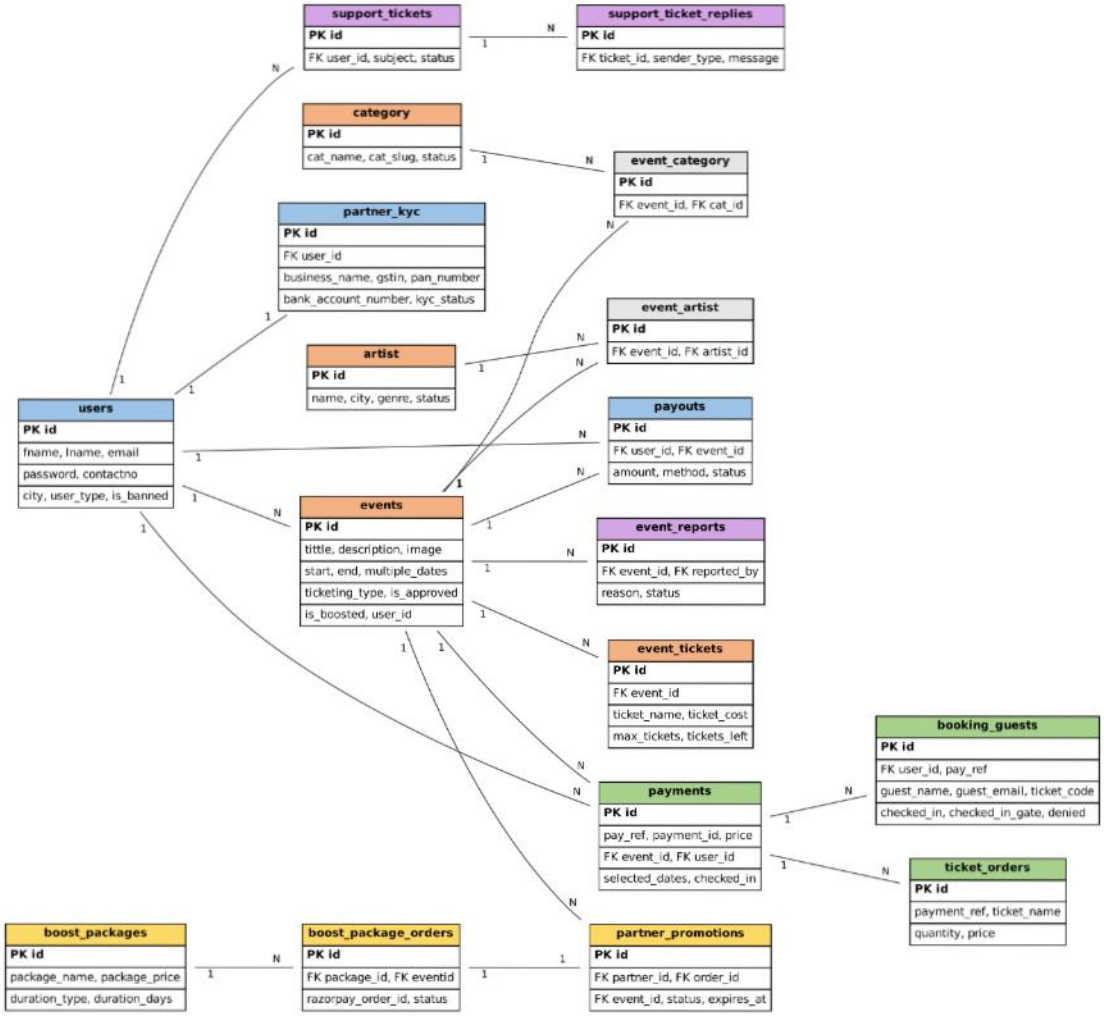


Level 1 DFD



ER Diagram

The ER diagram shows the main entities and their relationships. Colours group the tables: accounts and payouts (blue), events and master data (orange), bookings and payments (green), promotions (yellow) and support (purple). The full database has 35 tables.



Database Design

The main tables of the MySQL database are described below. The database has 35 tables in total; the remaining tables store users' KYC and payout details, promotions, support tickets, reports, notifications, site content and location master data.

1. users

Field	Data type (size)	Constraints	Description
id	Int(11)	Primary key	Unique record id
fname	Varchar(255)	Null	First name
lname	Varchar(255)	Null	Last name
gender	Varchar(225)	Not null	Gender
city	Varchar(255)	Not null	City
status	Int(11)	Not null (default 0)	Record status
is_banned	Tinyint(1)	Not null (default 0)	Banned flag (1 = banned)
email	Varchar(255)	Not null	Email address
password	Varchar(300)	Null	Encrypted password
contactno	Varchar(11)	Null	Contact number
image_url	Varchar(299)	Not null	Profile / image URL
posting_date	Timestamp	Not null	Registration date and time
user_type	Int(99)	Not null (default 0)	Type of account / role
reset_token	Varchar(255)	Null	Password reset token
reset_expiry	Datetime	Null	Reset token expiry
updated_at	Timestamp	Null	Last update time

2. events

Field	Data type (size)	Constraints	Description
id	Int(11)	Primary key	Unique record id
cat_id	Text	Not null (default 0)	Id of the category
tittle	Text	Not null	Event title
sub_tittle	Text	Not null	Event sub-title
description	Text	Not null	Description
term	Text	Not null	Terms and conditions
video	Text	Not null (default #)	Promo video link
image	Text	Not null	Image
Organised	Text	Not null	Organiser name
special_detail	Text	Not null	Special details
start	Datetime	Not null	Start date and time
end	Datetime	Not null	End date and time
multiple_dates	Text	Not null	List of event dates
multiple_date_id	Int(99)	Not null	Stores multiple date id
views_count	Int(255)	Null (default 0)	Number of views
url	Text	Not null	URL slug
event_status	Int(99)	Not null (default 0)	Stores event status
banner_status	Int(99)	Not null (default 0)	Stores banner status
status	Int(11)	Null (default 0)	Record status
ticketing_type	Varchar(299)	Not null	Ticketing type
user_id	Varchar(299)	Not null	Id of the related user
is_approved	Int(11)	Null (default 0)	Admin approval flag
is_boosted	Int(11)	Null (default 0)	Boosted flag
boost_package_id	Int(11)	Null (default 0)	Active boost package
created_at	Varchar(555)	Null	Creation time

3. event_tickets

Field	Data type (size)	Constraints	Description
id	Int(11)	Primary key	Unique record id
event_id	Int(11)	Not null	Id of the related event
created_by	Int(20)	Not null	Partner who created the ticket
ticket_name	Varchar(255)	Not null	Ticket type name
max_tickets	Int(11)	Not null	Total tickets available
tickets_left	Int(11)	Not null	Tickets remaining
ticket_cost	Varchar(299)	Not null	Price per ticket
ticket_desc	Text	Null	Ticket description
created_at	Timestamp	Not null	Creation time

4. payments

Field	Data type (size)	Constraints	Description
id	Int(99)	Primary key	Unique record id
pay_ref	Varchar(299)	Not null	Payment / booking reference
name	Text	Not null	Name
email	Text	Not null	Email address
phone	Text	Not null	Phone number
event_id	Text	Not null	Id of the related event
selected_dates	Text	Not null	Selected event date(s)
price	Varchar(299)	Not null	Amount paid
payment_id	Text	Not null	Razorpay payment id
created_at	Datetime	Not null	Creation time
user_id	Int(10)	Null	Id of the related user
gender	Varchar(10)	Null	Gender
is_self	Tinyint(1)	Not null (default 1)	1 if booked for self
checked_in	Tinyint(1)	Not null (default 0)	Entry check-in flag
checked_in_at	Datetime	Null	Check-in time
checked_in_gate	Varchar(50)	Null	Gate used for entry
denied	Tinyint(1)	Not null (default 0)	Entry denied flag
denied_reason	Varchar(500)	Null	Reason for denial
denied_at	Datetime	Null	Time of denial
denied_by	Int(11)	Null	Staff who denied entry

5. booking_guests

Field	Data type (size)	Constraints	Description
id	Int(10)	Primary key	Unique record id
user_id	Int(10)	Not null	Id of the related user
pay_ref	Varchar(64)	Not null	Payment / booking reference
is_self	Tinyint(1)	Not null (default 1)	1 if booked for self
guest_name	Varchar(150)	Not null	Guest name
guest_email	Varchar(150)	Not null	Guest email
guest_phone	Varchar(15)	Not null	Guest phone
guest_gender	Varchar(10)	Null	Guest gender
checked_in	Tinyint(1)	Not null (default 0)	Entry check-in flag
checked_in_at	Datetime	Null	Check-in time
checked_in_gate	Varchar(50)	Null	Gate used for entry
checked_in_by	Int(11)	Null	Staff who checked in
denied	Tinyint(1)	Not null (default 0)	Entry denied flag
denied_reason	Varchar(500)	Null	Reason for denial
denied_at	Datetime	Null	Time of denial
denied_by	Int(11)	Null	Staff who denied entry
ticket_code	Varchar(64)	Null	Unique ticket / QR code
created_at	Timestamp	Null	Creation time

6. boost_packages

Field	Data type (size)	Constraints	Description
id	Int(11)	Primary key	Unique record id
user_id	Int(11)	Not null (default 0)	Id of the related user
admin_user_id	Int(11)	Null	Admin who created the package
package_name	Varchar(555)	Null	Package name
package_description	Text	Not null	Package description
package_price	Varchar(255)	Null	Package price
duration_type	Enum	Not null (default fixed)	Fixed days or until event end: fixed, until_event_end
duration_days	Int(11)	Null	Number of days
allow_multiple_dates	Tinyint(1)	Not null (default 0)	Allow several event dates
status	Tinyint(1)	Not null (default 1)	Record status
created_at	Varchar(255)	Null	Creation time
updated_at	Varchar(255)	Null	Last update time

7. payouts

Field	Data type (size)	Constraints	Description
id	Int(11)	Primary key	Unique record id
user_id	Int(11)	Not null	Id of the related user
event_id	Int(11)	Not null	Id of the related event
amount	Decimal(12,2)	Not null	Payout amount
method	Enum	Not null	Payout method (bank / UPI): bank, upi
status	Enum	Not null (default pending)	Record status: pending, processing, paid, failed
reference_id	Varchar(100)	Null	Bank / UPI reference
admin_notes	Text	Null	Admin remarks
requested_at	Timestamp	Not null	Request time
processed_at	Timestamp	Null	Processing time
processed_by	Int(11)	Null	Admin who processed

Modules

The system has four modules: Administrator (controls the entire platform), Partner / Event Organizer (creates events and manages entry), User (books tickets) and Guest (browses events without logging in).

Functional Specification

Administrator

a) Manage user and partner accounts, including banning accounts.
b) Approve or reject events; verify partner KYC details.
c) Manage categories, artists, places, banners, blogs and pages.
d) Create boost packages; process payouts to partners.
e) Respond to support tickets and resolve event reports.

Partner (Event Organizer)

a) Register, log in and submit KYC and payout details.
b) Create and edit events and define ticket types with price and quantity.
c) View bookings and guest lists; check in or deny guests using the ticket code.
d) Buy boost packages, request payouts, raise support tickets.

User

a) Register, log in and update profile; search events by city, category, artist and place.
b) Book tickets for self and guests and pay online through Razorpay.
c) Receive SMS and e-ticket with a unique ticket code; view booking history.
d) Report an event or raise a support ticket.

Guest

a) View events, event details, artists, places and blogs; register or log in to book.

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